

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 5. Conditional Statements

Lecture: 40. Challenge - Eligible to Work

Section 1: Lecture Summary

This lecture focused on writing and understanding compound conditional statements in Python through solving three practical challenges. The instructor first explained the problem of determining eligibility to work based on age, emphasizing the need to check if a person's age is between 18 and 60 inclusive using compound conditions combined with the logical "and" operator. Next, the challenge of validating marks for an exam was introduced, requiring validation that marks are within 0 to 100. This demonstrated how to apply compound conditions for numeric ranges. Lastly, the lecture covered gender checking where input could be lowercase or uppercase letters, illustrating how to check multiple string equality conditions using the logical "or" operator. Throughout, simple Python programs were constructed using if-else logic to solve these problems, reinforcing the syntax and logic of compound conditional statements.

Section 2: Key Concepts and Explanations

- Compound Conditional Statements
- Combine multiple boolean expressions using logical operators such as:
 - **and**: Both conditions must be true.
 - **or**: At least one condition must be true.
- Syntax: ``if condition1 and condition2:``
- Example: To check if age is in a certain range (18 to 60),

eligible

```
if age >= 18 and age <= 60:
```

- Eligibility Check Based on Age

- Eligibility criteria: Age must be ≥ 18 and ≤ 60 .
- Both conditions must be true for eligibility.
- Use `and` operator.

- Marks Validity Check

- Valid marks must be between 0 and 100 inclusive.
- Conditions:
 - marks ≥ 0
 - marks ≤ 100
- Both must be true; use `and`.
- Invalid if marks are negative or exceed 100.

- Gender Check

- Accepts gender input as a single character.
- Male can be represented as 'm' or 'M'.
- Female is assumed otherwise.
- Use `or` operator to check for 'm' or 'M':

```
if gender == 'm' or gender == 'M':  
    print('Male')  
else:  
    print('Female')
```

- Input Handling and Print Output

- Use `input()` function to accept user input.
- Convert input string to integer using `int()` for numeric checks.
- Print appropriate messages based on condition results.

Section 3: Example Code and Use Cases

1. Age Check for Work

```
age = int(input("Enter your Age: "))
if age >= 18 and age <= 60:
    print("Can Work")
else:
    print("Cannot Work")
```

This program asks the user for their age and prints whether the person is eligible to work based on the age range 18 to 60.

2. Valid Marks Check

```
marks = int(input("Enter your Marks: "))
if marks >= 0 and marks <= 100:
    print("Valid Marks")
else:
    print("Invalid Marks")
```

This program requests the marks obtained in a subject and verifies if the marks are within the valid range from 0 to 100.

3. Gender Check

```
gender = input("Enter your Gender (M/F): ")
if gender == 'm' or gender == 'M':
    print("Male")
else:
    print("Female")
```

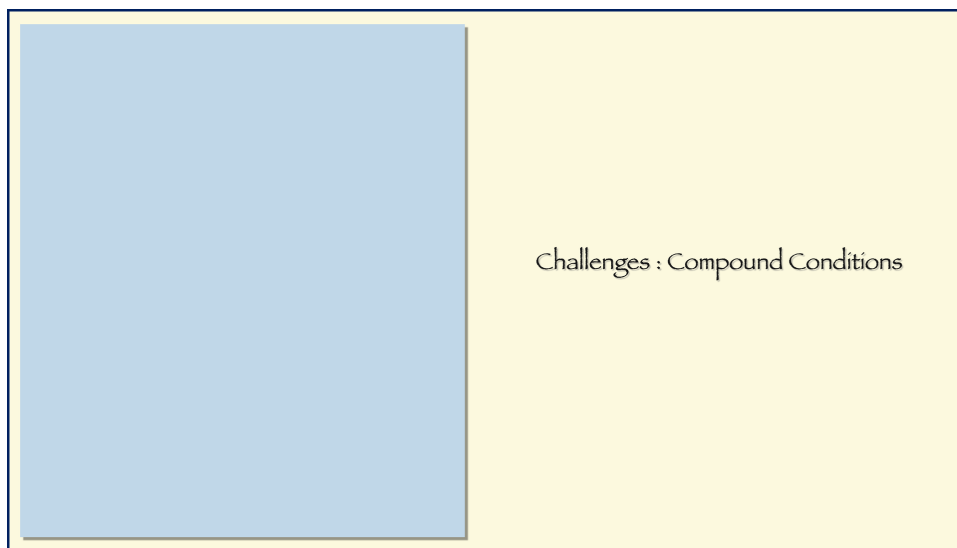
This program takes a gender input and identifies the gender as Male if the input is either lowercase or uppercase 'm'; otherwise, it prints Female.

Section 4: Key Takeaways

- Compound conditional statements combine multiple conditions using logical operators.
- The `and` operator requires all conditions to be true for the overall condition to be true.
- The `or` operator requires at least one condition to be true.
- Always use `:` after the if or else conditions and indent the code block properly.
- Convert input strings to integers for numeric comparisons with `int()`.
- Simple Python programs can perform practical checks with only a few lines of code.
- Consistency in condition writing (`>=`, `<=`) and logical operator usage is essential for correctness.
- Case sensitivity matters when comparing strings; check for both uppercase and lowercase if needed.
- These foundational conditional constructs are critical for validating inputs and decision-making in programs.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Challenges : Compound Conditions

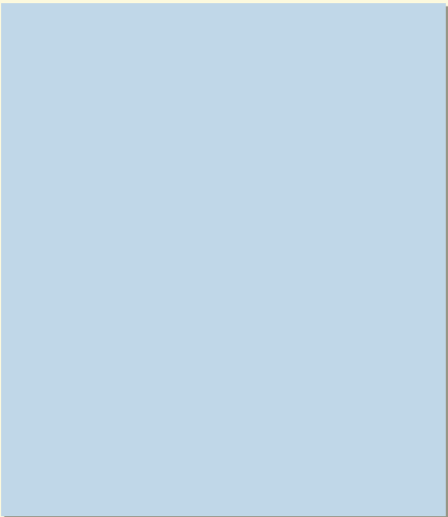
- Age Check for Work

Challenges : Compound Conditions

- Age Check for Work
- Valid Marks

Challenges : Compound Conditions

- Age Check for Work
- Valid Marks
- Gender Check

- 
- Age Check for Work

Age Check for Work
Challenge

Age Check for Work
Challenge



Age Check for Work
Challenge



Age Check for Work
Challenge



Age Check for Work
Challenge



Age Check for Work
Challenge



age < 18

Age Check for Work

Challenge



age < 18





age > 60

Age Check for Work

Challenge



age < 18



age = 20



age > 60

Age Check for Work

Challenge



age < 18



age = 20



age > 60

age >= 18

Age Check for Work

Challenge


age < 18


age = 20


age > 60

age >= 18

age <= 60

Age Check for Work

Challenge


age < 18


age = 20


age > 60

age >= 18 and age <= 60

Age Check for Work

Challenge



age < 18



age = 20



age > 60

age >= 18 and age <= 60

Age Check for Work

Challenge



age < 18



age = 20



age > 60

if (age >= 18 and age <= 60):

Age Check for Work

Challenge



age < 18



age = 20



age > 60

```
if (age >= 18 and age <= 60):  
    print( 'Can Work' )
```

Age Check for Work

Challenge



age < 18



age = 20



age > 60

```
if (age >= 18 and age <= 60):  
    print( 'Can Work' )  
else:
```

Age Check for Work

Challenge



age < 18



age = 20



age > 60

```
if (age >= 18 and age <= 60):  
    print( 'Can Work' )  
else:  
    print( 'Cannot Work' )
```

Age Check for Work

Challenge



age < 18



age = 20



age > 60

```
if (age >= 18 and age <= 60):  
    print( 'Can Work' )  
else:  
    print( 'Cannot Work' )
```

Age Check for Work Challenge



age = ?

```
if (age >= 18 and age <= 60):  
    print( 'Can Work' )  
else:  
    print( 'Cannot Work' )
```

Age Check for Work

Challenge



age < 18

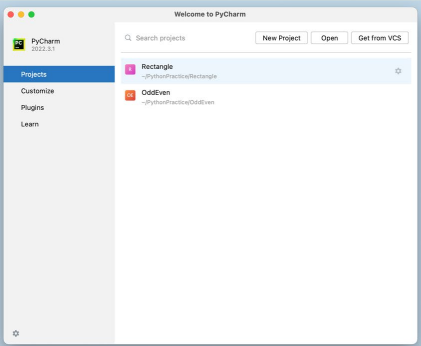


age = 20



age > 60

```
age = int( input( 'Enter your Age' ) )
if ( age >= 18 and age <= 60 ):
    print( 'Can Work' )
else:
    print( 'Cannot Work' )
```



Age Check for Work

Challenge



age < 18



age = 20



age > 60

```
age = int( input( 'Enter your Age' ) )
if ( age >= 18 and age <= 60 ):
    print( 'Can Work' )
else:
    print( 'Cannot Work' )
```

Location: /Users/abdubar/PythonPractice/

Python Interpreter: New Virtualenv environment

New environment using Virtualenv

Location: /Users/abdubar/PythonPractice/

Base Interpreter: Python 3.11 JupyterLabPython3.11

☐ Inherit global site-packages

☐ Make available to all projects

☐ Previously configured interpreter

☒ Create a main.py welcome script

Create a Python script that provides an entry point to coding in PyCharm.

Environment location directory is not empty

Cancel Create

Age Check for Work

Challenge



age < 18



age = 20



age > 60

```
age = int( input( 'Enter your Age' ) )  
if ( age >= 18 and age <= 60 ):  
    print( 'Can Work' )  
else:  
    print( 'Cannot Work' )
```

Location: /Users/abdubar/PythonPractice/WorkAge/

Python Interpreter: New Virtualenv environment

New environment using Virtualenv

Location: /Users/abdubar/WorkAge/

Base Interpreter: Python 3.11 JupyterLabPython3.11

☐ Inherit global site-packages

☐ Make available to all projects

☐ Previously configured interpreter

☒ Create a main.py welcome script

Create a Python script that provides an entry point to coding in PyCharm.

Cancel Create

Age Check for Work

Challenge



age < 18



age = 20



age > 60

```
age = int( input( 'Enter your Age' ) )  
if ( age >= 18 and age <= 60 ):  
    print( 'Can Work' )  
else:  
    print( 'Cannot Work' )
```

WorkAge - main.py

main.py

Age Check for Work

Challenge



```
age = int( input( 'Enter your Age' ) )  
if ( age >= 18 and age <= 60 ):  
    print( 'Can Work' )  
else:  
    print( 'Cannot Work' )
```

WorkAge - main.py

main.py

1
2
3
4
5
6
7

age = int(input('Enter Age'))

if age >= 18 and age <= 60:
 print('Can Work')
else:
 print('Cannot Work')

Age Check for Work

Challenge



```
age = int( input( 'Enter your Age' ) )  
if ( age >= 18 and age <= 60 ):  
    print( 'Can Work' )  
else:  
    print( 'Cannot Work' )
```


WorkAge - main.py

```
1 age = int(input('Enter Age: '))
2
3 if age >= 18 and age <= 60:
4     print('Can Work')
5 else:
6     print('Cannot Work')
7
```

Run: main
/Users/abdulbari/WorkAge/bin/python /Users/ab
Enter Age:

Age Check for Work

Challenge



age < 18



age = 20



age > 60

```
age = int( input( 'Enter your Age' ))
if (age >= 18 and age <= 60):
    print( 'Can Work' )
else:
    print( 'Cannot Work' )
```

WorkAge - main.py

```
1 age = int(input('Enter Age: '))
2
3 if age >= 18 and age <= 60:
4     print('Can Work')
5 else:
6     print('Cannot Work')
7
```

Run: main
/Users/abdulbari/WorkAge/bin/python /Users/ab
Enter Age: 20
Can Work

Process finished with exit code 0

Age Check for Work

Challenge



age < 18

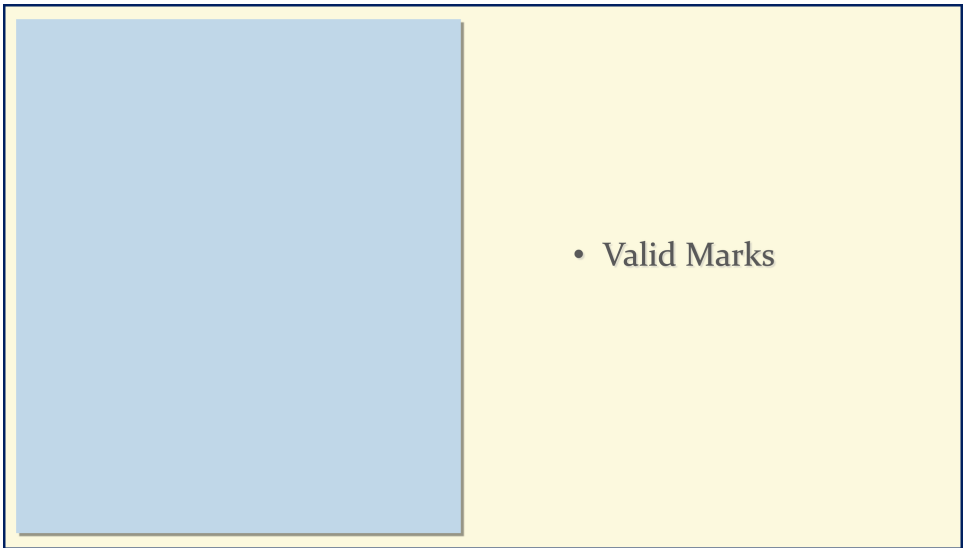
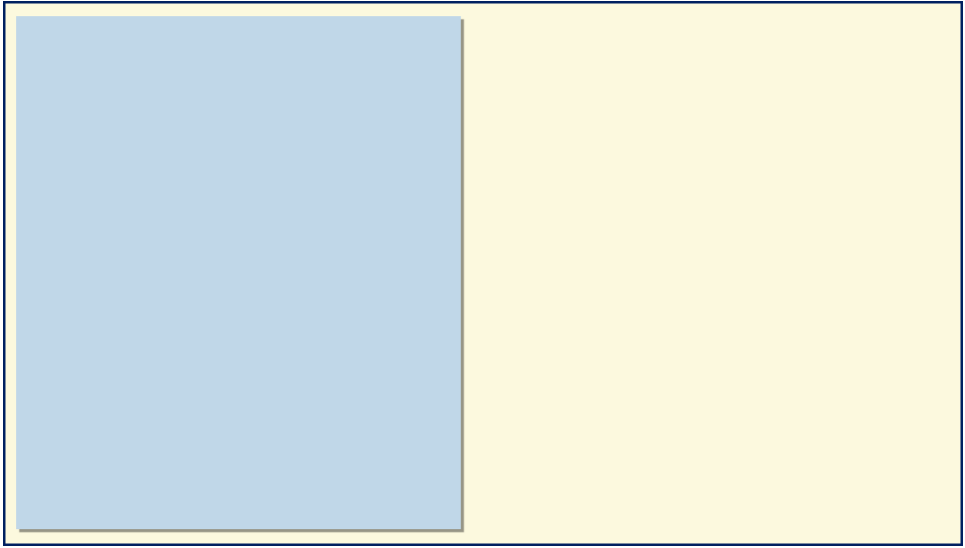


age = 20



age > 60

```
age = int( input( 'Enter your Age' ))
if (age >= 18 and age <= 60):
    print( 'Can Work' )
else:
    print( 'Cannot Work' )
```



Valid Marks Challenge

The diagram illustrates a 'Valid Marks Challenge' process. It begins with a large blue rectangle on the left, representing the initial state. An arrow points from this rectangle to a yellow rectangle on the right, which contains the text 'Valid Marks Challenge'. This yellow rectangle is part of a larger yellow area that also contains a smaller yellow rectangle labeled 'Challenge' with an arrow pointing to it. The entire process is enclosed in a black border.

Valid Marks

Challenge



Valid Marks

Challenge

Max Marks =100



Valid Marks

Challenge

Max Marks =100



Marks

0 — 100

Valid Marks

Challenge

Max Marks =100



Marks

0 — 100

marks >= 0

Valid Marks

Challenge

Max Marks =100



Marks

0 — 100

marks >= 0

marks <= 100

Valid Marks

Challenge

Max Marks =100



Marks

0 — 100

marks >= 0 and marks <= 100

Valid Marks

Challenge

Max Marks =100



Marks
0 — 100

marks >= 0 and marks <= 100

Valid Marks

Challenge

Max Marks =100



Marks
0 — 100

if (marks >= 0 and marks <= 100) :

Valid Marks

Challenge

Max Marks =100



Marks
0 — 100

```
if (marks >= 0 and marks <= 100):  
    print( 'Valid Marks' )
```

Valid Marks

Challenge

Max Marks =100



Marks
0 — 100

```
if (marks >= 0 and marks <= 100):  
    print( 'Valid Marks' )  
else:
```


Valid Marks

Challenge

Max Marks =100



Marks
0 — 100

```
if (marks >= 0 and marks <= 100):  
    print( 'Valid Marks' )  
else:  
    print( 'inValid Marks' )
```

Valid Marks

Challenge

Max Marks =100



```
if ( marks >= 0  and  marks <= 100 ):  
    print( 'Valid Marks' )  
else:  
    print( 'inValid Marks' )
```

Valid Marks

Challenge

Max Marks =100



```
marks = ?  
  
if ( marks >= 0  and  marks <= 100 ):  
    print( 'Valid Marks' )  
else:  
    print( 'inValid Marks' )
```

Valid Marks

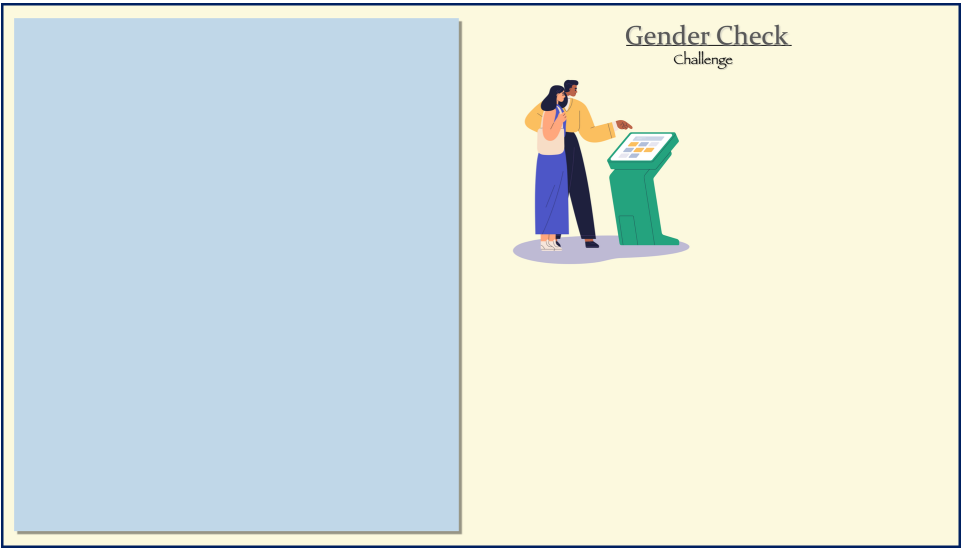
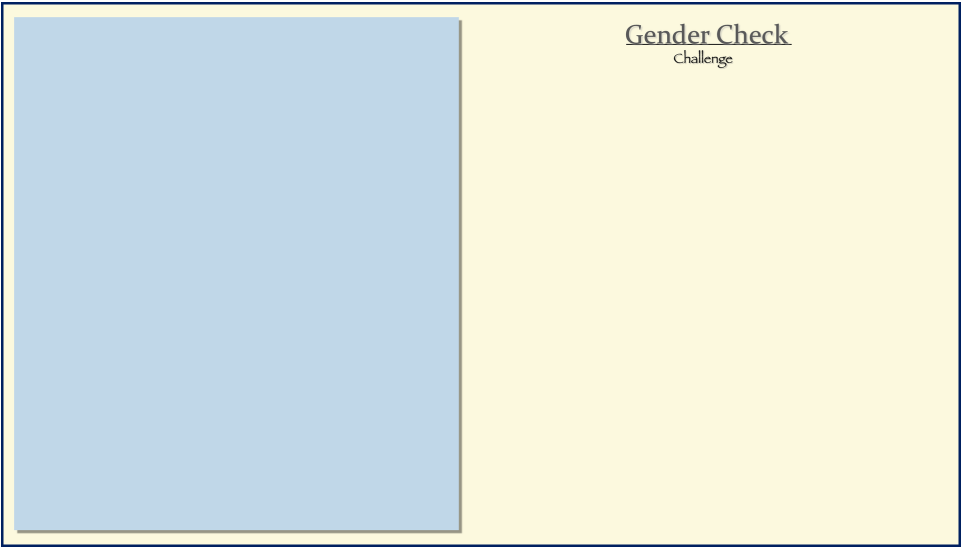
Challenge

Max Marks =100



```
marks = int( input( 'Enter your Marks' ))  
  
if ( marks >= 0  and  marks <= 100 ) :  
    print( 'Valid Marks' )  
else:  
    print( 'inValid Marks' )
```

- Gender Check



Gender Check

Challenge



Gender Check

Challenge



gender = ?

Gender Check

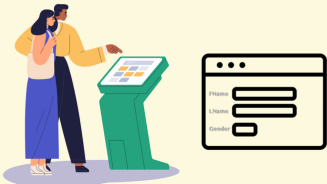
Challenge



gender = m gender = M

Gender Check

Challenge



gender = m gender = M
gender = f gender = F

Gender Check Challenge



First Name	
Last Name	
Gender	

`gender = m` `gender = M`

`gender == 'm'`

Gender Check Challenge



`gender = m    gender = M`

`gender == 'm'` `gender == 'M'`

Gender Check Challenge



`gender = m    gender = M`

`gender == 'm'   or   gender == 'M'`

Gender Check

Challenge



`gender = m` `gender = M`

```
if (gender == 'm' or gender == 'M'):
```

Gender Check

Challenge



`gender = m` `gender = M`

```
if (gender == 'm' or gender == 'M'):  
    print('Male')
```

Gender Check

Challenge



gender = m gender = M

```
if (gender == 'm' or gender == 'M'):
    print( 'Male' )
else:
```

Gender Check

Challenge



gender = m gender = M

```
if (gender == 'm' or gender == 'M'):
    print( 'Male' )
else:
    print( 'Female' )
```

Gender Check

Challenge



gender = m gender = M

```
if (gender == 'm' or gender == 'M'):  
    print( 'Male' )  
else:  
    print( 'Female' )
```

Gender Check

Challenge



gender = m gender = M

```
gender = ?  
if (gender == 'm' or gender == 'M'):  
    print( 'Male' )  
else:  
    print( 'Female' )
```

Gender Check Challenge



gender = m gender = M


```
gender = int( input( 'Enter your Gender' ) )  
if( gender == 'm' or gender == 'M' ):  
    print( 'Male' )  
else:  
    print( 'Female' )
```